

Why A1C Is Clinically Meaningful and a Good Severity Proxy

What A1C Measures

A1C (HbA1c) reflects a patient's average blood glucose level over the past 2–3 months. It measures how much glucose is bound to hemoglobin in red blood cells, providing a long-term view of glycemic control rather than a single snapshot.

Clinical Interpretation

Lower A1C values indicate good glucose control, while higher values indicate poor control. Higher A1C levels are associated with increased risk of complications, infections, delayed healing, and medication-related issues.

Why A1C Matters for Readmissions

Patients with poorly controlled diabetes are more likely to experience complications that increase hospital length of stay and the probability of readmission. A1C therefore provides insight into a patient's baseline health status prior to hospitalization.

What “Severity Proxy” Means

A proxy is a variable that approximates an underlying concept that cannot be measured directly. In this dataset, we lack explicit measures of disease stage, frailty, or complication burden. A1C serves as a proxy for chronic disease severity and quality of outpatient management.

Why Missing A1C Is Informative

An A1C value that is not measured does not mean normal control. It may indicate that diabetes was not the focus of care or that clinicians did not suspect poor control. Preserving this missingness allows models to learn meaningful clinical patterns.

Interview-Ready Summary

A1C reflects long-term glycemic control and serves as a clinically grounded proxy for chronic disease severity. Missing A1C values are treated as informative rather than imputed, preserving real-world clinical context.